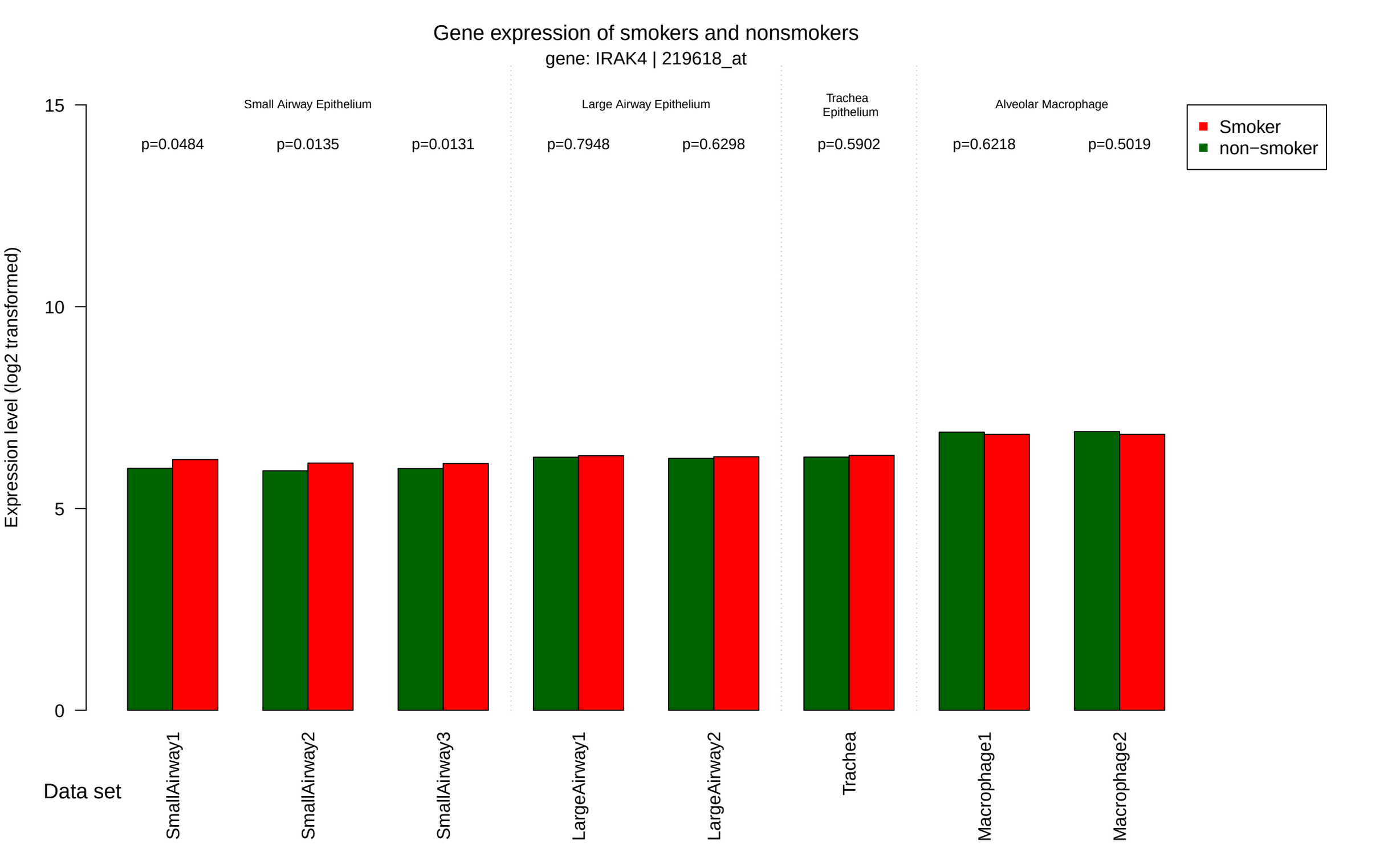
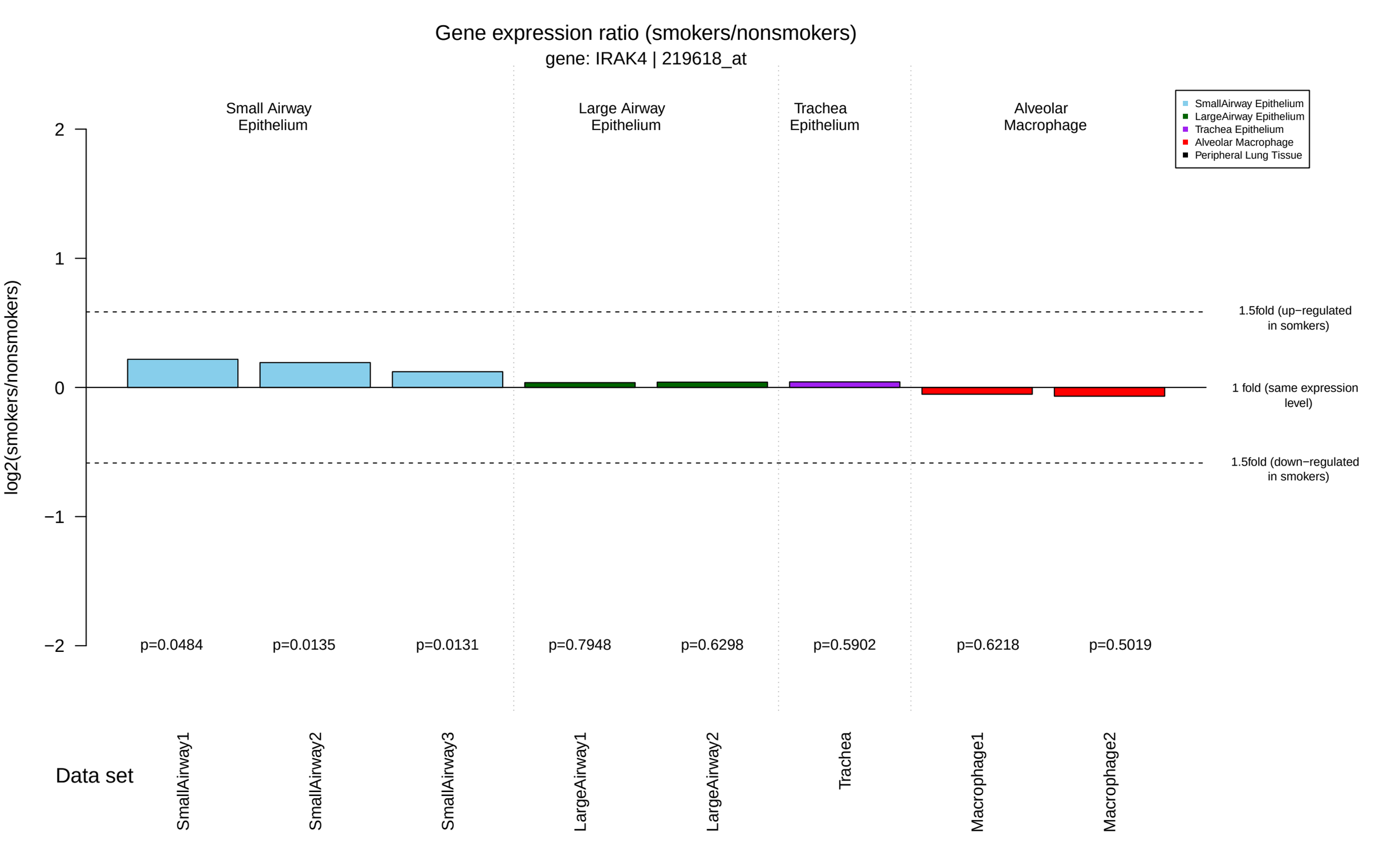
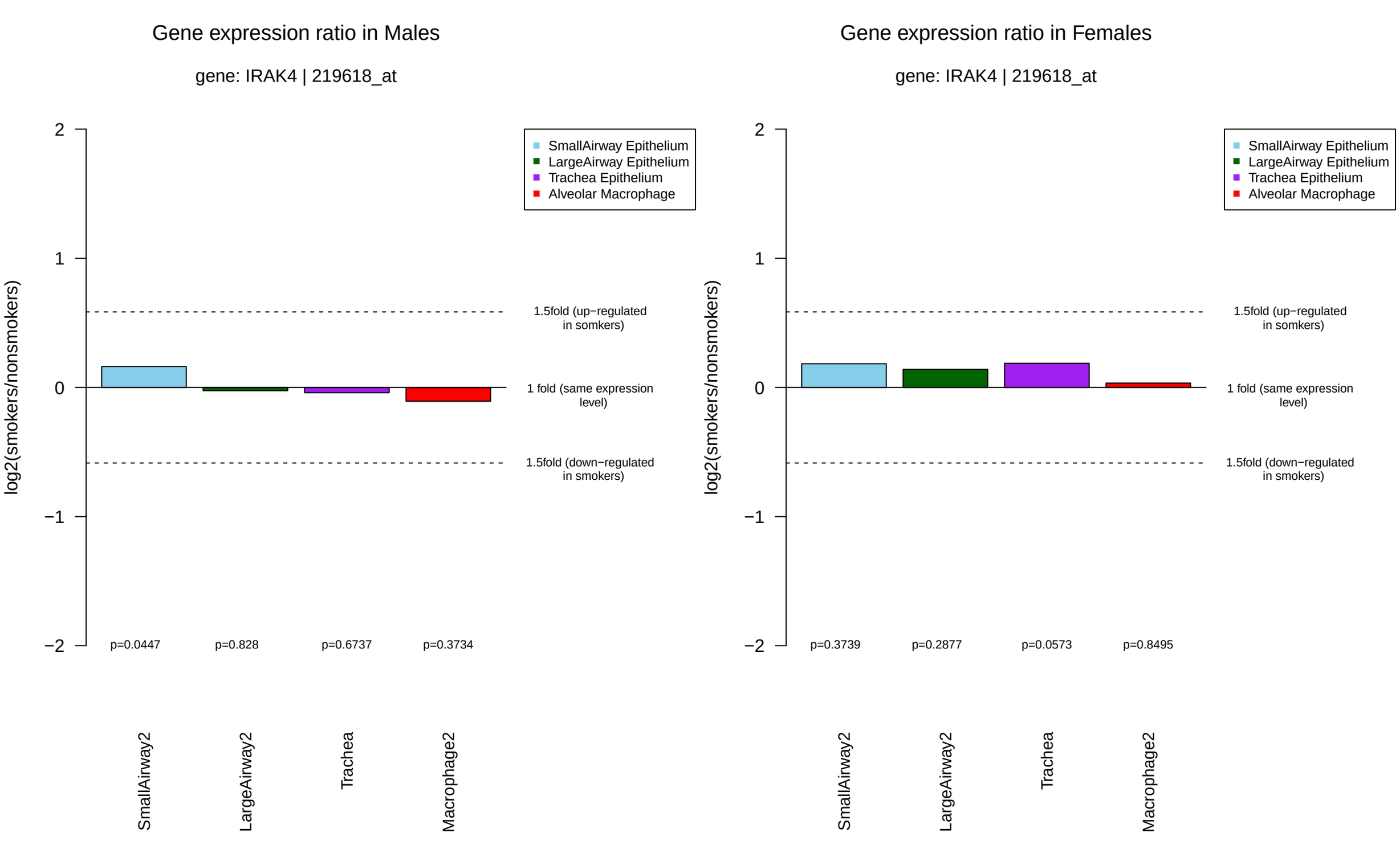
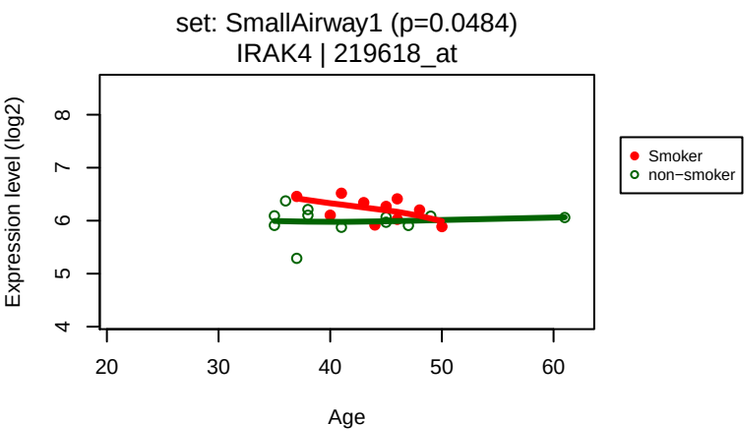
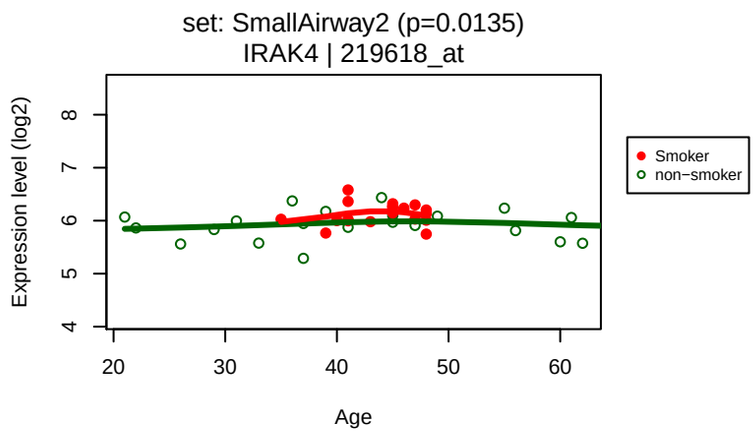
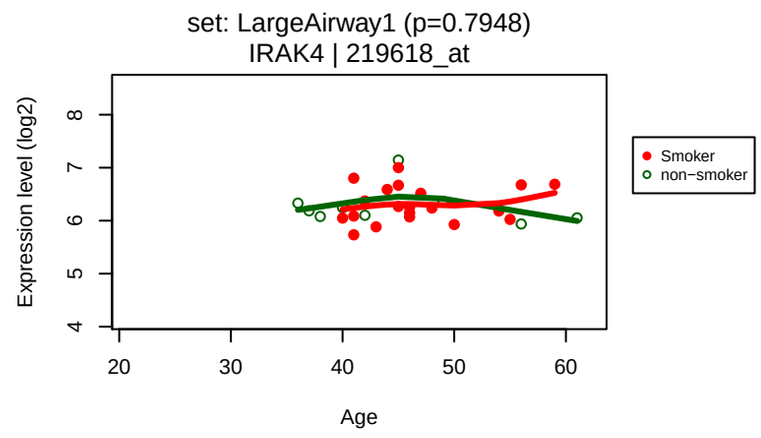

Gene expression of smokers and nonsmokers
gene: IRAK4 | 219618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0484
p=0.0135
p=0.0131
p=0.7948
p=0.6298
p=0.5902
p=0.6218
p=0.5019
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

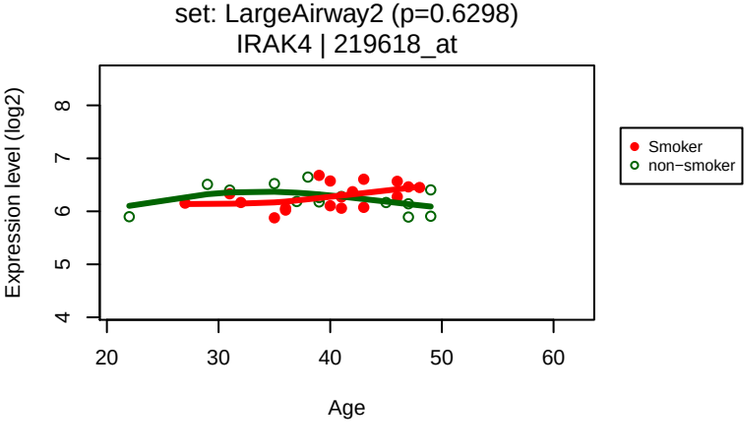

Gene expression ratio (smokers/nonsmokers)
gene: IRAK4 | 219618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0484
p=0.0135
p=0.0131
p=0.7948
p=0.6298
p=0.5902
p=0.6218
p=0.5019
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

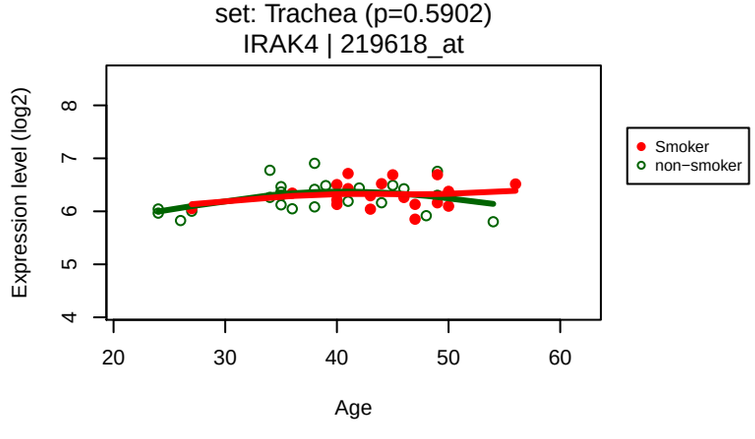

Gene expression ratio in Males
gene: IRAK4 | 219618_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0447
p=0.828
p=0.6737
p=0.3734
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRAK4 | 219618_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3739
p=0.2877
p=0.0573
p=0.8495
SmallAirway2
LargeAirway2
Trachea
Macrophage2

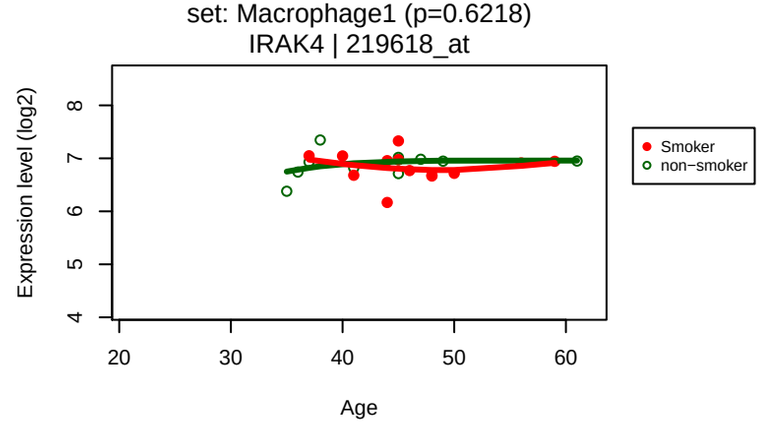

set: SmallAirway1 (p=0.0484)
IRAK4 | 219618_at
Expression level (log2)
Age
Smoker
non−smoker

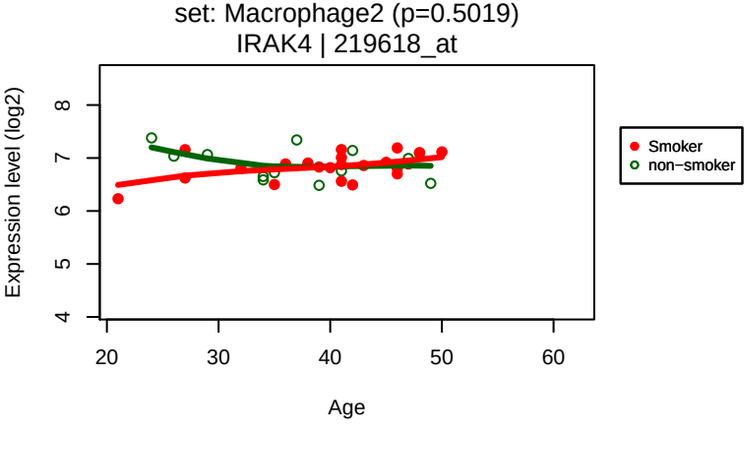

set: SmallAirway2 (p=0.0135)
IRAK4 | 219618_at
Expression level (log2)
Age
Smoker
non−smoker

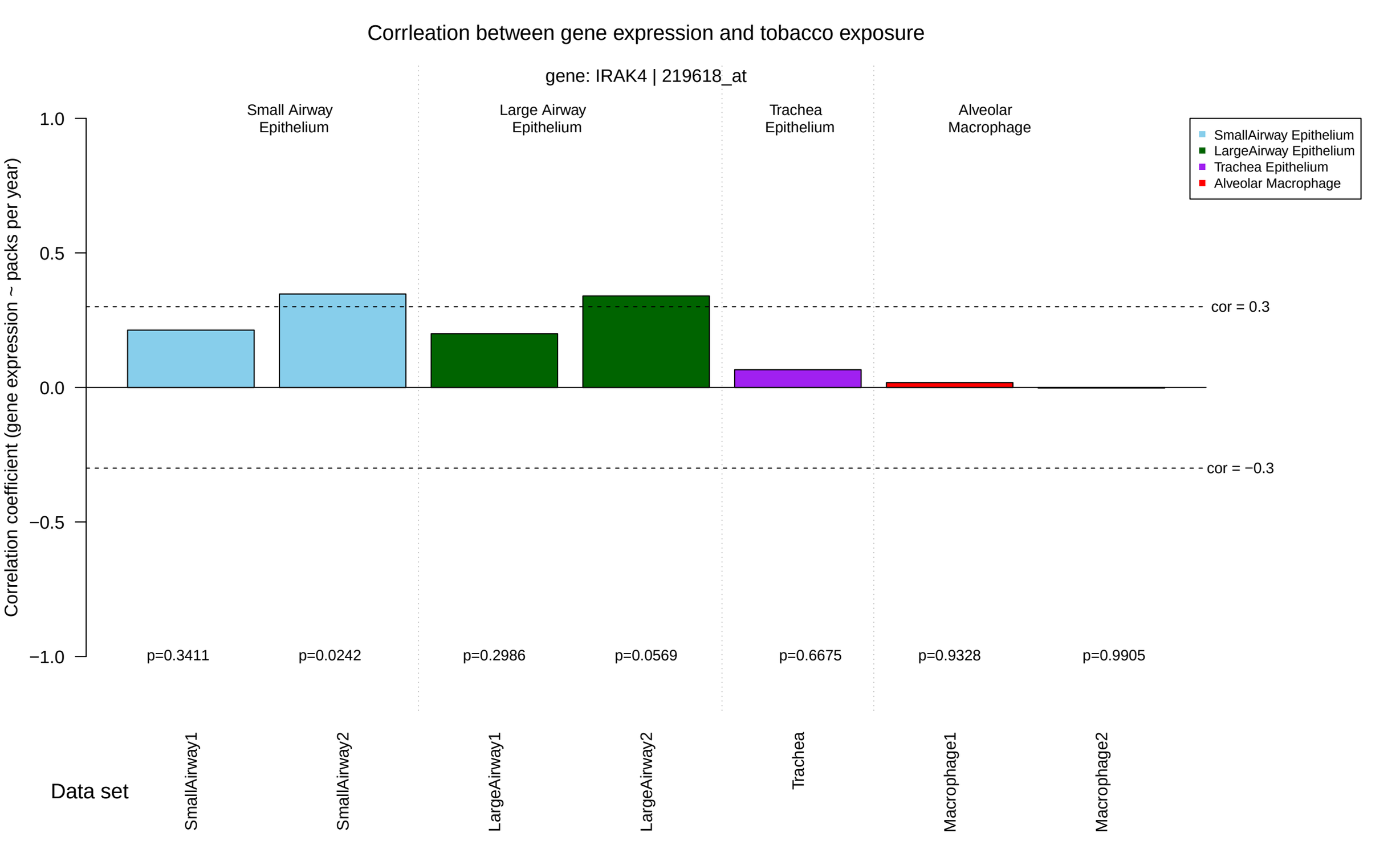

set: LargeAirway1 (p=0.7948)
IRAK4 | 219618_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6298)
IRAK4 | 219618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5902)
IRAK4 | 219618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6218)
IRAK4 | 219618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5019)
IRAK4 | 219618_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRAK4 | 219618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3411
p=0.0242
p=0.2986
p=0.0569
p=0.6675
p=0.9328
p=0.9905
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2